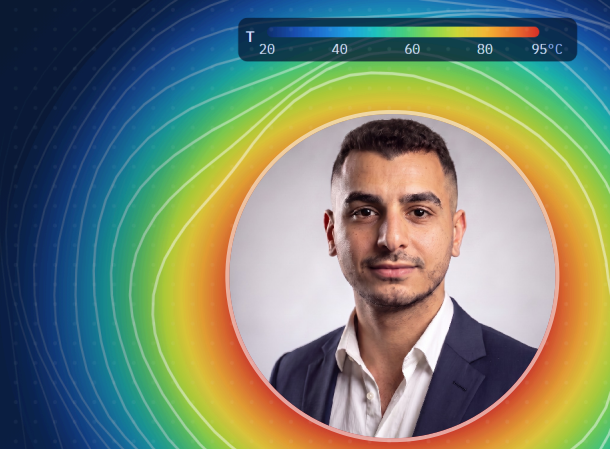
CURRICULUM VITAE · 2026

KHALED ABDELHADI

Thermal & Fluid Engineer for vehicle cooling, hydraulics and heat transfer.

Mechanical Engineer (M.Eng.) · 1D/3D-CFD · Hydraulics · CAD & FEM

— Cooling systems proven in simulation before hardware is built.



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German / Egyptian Citizen Born 20 July 1995 (31) **German** Professional · **English** Fluent · **Arabic** Native

5 YEARS
Thermal & Fluid Simulation

3 DRIVETRAINS
Battery-Electric, Diesel & Hydrogen Fuel Cell

2 DEGREES
B.Sc. & M.Eng. in Mechanical Engineering

3 LANGUAGES
German, English, Arabic

PROFILE

Mechanical engineer (M.Eng.) with **five years in thermal management, hydraulics and simulation** at MAN Truck & Bus and TRATON R&D. I built and owned the **1D cooling-circuit models for electric, diesel and hydrogen fuel-cell trucks**, calibrated against 3D CFD and test-bench data and used for concept and release decisions. I size pumps, valves and pipes from pressure-loss calculations and turn the results into **design decisions**. I wrote my team's simulation methodology and built variant models reused across platforms. Also hands-on in 3D design, FEM and composites.

Thermal Management · 1D / 3D-CFD · GT-SUITE (expert) · Hydraulics · Battery & Fuel-Cell Cooling · FEM / FEA · 3D CAD Design · 2D Drawings & Tolerances · Manufacturing · Simulation Methodology · Python · MATLAB

EXPERTISE

Thermal
Thermal management, heat transfer, thermodynamics, cooling circuits (battery, e-motor, HV, cabin, refrigerant, hydrogen fuel cell), heat exchangers, pipe insulation, immersion cooling, transient drive-cycle simulation, extreme-ambient, worst-case and safety (FTTI) scenarios, thermal hotspot analysis

Hydraulic & Fluid
Hydraulic circuit design and modelling, pressure loss and pipe sizing calculations, flow rate and flow distribution analysis across interconnected circuits, pump sizing and operating points, hydraulic control elements (2-way and 3-way valves, check valves, thermostats, flow restrictors), venting and elevation design, hydraulic validation against measurements, P&ID reading, industrial piping (ASME B31.1 / B31.3 / B36, single- and two-phase pipe hydraulics, valves, flanges)

Simulation
1D / 3D-CFD system simulation (GT-SUITE, expert), FEM / FEA structural simulation (Altair HyperWorks; Abaqus / Nastran basic), electromagnetic simulation (ANSYS Maxwell), STAR-CCM+ (course project: external-flow geometry, boundary and mesh setup), plastic-flow simulation, model calibration and validation against test data, simulation methodology

CAD & FEM
3D part and assembly design, 2D drawings with tolerances (DIN / ISO), mechanical design of components, stress and stiffness assessment, structural optimization (weight, torsional stiffness), CFRP composites, BOM creation; CAD: PTC Creo, SpaceClaim

Manufacturing
Injection mold (steel tool) design, CNC programming, composite layup and curing, prototype build, supplier data evaluation, molding-defect elimination, manufacturing documentation

Tools & Methods
MATLAB/Simulink, Python (data analysis, automation), AUDI loss-map toolbox, JIRA, Agile / Scrum, V-model / ASPICE, failure analysis and root-cause investigation, trade-off studies, KPI tracking, technical documentation

EDUCATION

M.Eng. Design & Development in Mechanical & Automotive Engineering — Hochschule Esslingen, Germany · 2019 – 2021

B.Sc. Mechanical Engineering — Alfaisal University, Riyadh, Saudi Arabia · 2013 – 2018

EXPERIENCE

Self-Directed Study and Projects
Apr 2026 – Present · 6 months

Applied AI, Process Automation and Sales Fundamentals

- Learned to apply AI to engineering work with Claude and Python, and studied the fundamentals of sales; completed courses in AI for design and optimization, Python, Claude Code, industrial piping, computational fluid mechanics and STAR-CCM+.
- Built own automation tools, an interactive web CV and interview-preparation study sites.

1D Simulation Expert, Vehicle Cooling Systems
Apr 2025 – Mar 2026 · 1 year

TRATON R&D Germany GmbH, Nuremberg, Germany

1D Simulation Engineer, Vehicle Cooling Systems
May 2022 – Apr 2025 · 3 years

MAN Truck & Bus SE, Nuremberg, Germany

GT-SUITE · 1D / 3D-CFD · TEST-BENCH DATA · JIRA / AGILE

- Led thermal-management simulation for electric, diesel and fuel-cell truck cooling; owner of the 1D circuit models used for concept and release decisions.
- Built 1D hydraulic and thermal models from 3D pipe geometry and supplier data; calibrated coolers and circuits against 3D CFD and physical measurements with strong agreement.
- Performed pressure loss, flow rate, flow distribution and pipe sizing calculations across interconnected battery, HV-component, cabin and refrigerant circuits; sized pumps, operating points and hydraulic control elements (2-way and 3-way valves, check valves, restrictors), improving energy efficiency.
- Modelled the refrigerant and connected coolant circuits as one system, optimizing their interaction for lower power consumption.
- Ran worst-case, extreme-ambient and safety scenarios (incl. FTTI) and derived design changes: venting positions, flow restrictors, pipe diameters and elevations.
- Created configurable variant sub-models (e.g. battery count) reused across vehicle platforms, and wrote the team's simulation methodology, cutting onboarding time and standardizing model quality.
- Covered the cooling side of hydrogen fuel-cell drivetrains; included pipe insulation in coolant-line models and assessed immersion liquid cooling for a customer's battery-electric train with about 100 battery modules, comparing cell capacities, energy density and cooling performance.
- Analysed cooling failures and supported supplier component selection with simulation-based trade-off studies.

1D/2D Simulation Engineer, E-Motor Cooling
Dec 2021 – May 2022 · 6 months

MAN Truck & Bus SE, Nuremberg, Germany

ANSYS MAXWELL · AUDI LOSS-MAP TOOLBOX · GT-SUITE

- Simulated e-motor electromagnetic behaviour and losses; validated torque/speed against measurements and generated loss maps for the thermal model.
- Calibrated the GT thermal model to sensor data, achieving temperature predictions within measurement tolerance.
- Identified thermal hotspots, demagnetization risk and previously unaccounted system losses before hardware testing.

1D Simulation Student, E-Motor Cooling
Oct 2020 – Mar 2021 · 6 months

MAN Truck & Bus SE, Nuremberg, Germany

- Built a bus e-motor 1D thermal model in GT-SUITE from scratch, including oil-circuit design with conduction and convection paths; became the base model for all later e-motor cooling studies.
- Converted 3D geometry to 1D and validated transient thermal simulation against drive-cycle test data.

Structural Simulation Engineer, CFRP Monocoque (Formula Student)
Mar 2020 – Aug 2020 · 6 months

Esslingen Rennstall, Esslingen, Germany

ALTAIR HYPERWORKS · FEM · CFRP

- Optimized the CFRP monocoque of the Stallardo 19 race car with FEM for lower weight and higher torsional stiffness; delivered the production-ready BOM and 2D schematics for the Stallardo 20 chassis.

Engineering Development & Management Intern
Jan 2020 – Aug 2020 · 8 months

A123 Systems GmbH, Stuttgart, Germany

- Built project KPIs from JIRA tickets (V-model, ASPICE) for milestone forecasting; analysed BMS architecture, battery design and safety testing.

Mold Design & Manufacturing Intern
Jul 2017 – Jan 2018 · 7 months

Alfanar Electric, Riyadh, Saudi Arabia

PTC CREO · CNC · PLASTIC-FLOW SIMULATION

- Designed steel injection molds (3D assemblies, 2D drawings), programmed CNC machining and ran plastic-flow simulations to eliminate molding defects.

CFRP Design & Manufacturing Member
Jun 2016 – Jun 2018 · 2 years

Shell Eco-Marathon Asia, Singapore

- Designed the vehicle exterior in Creo, built foam molds, led carbon-fibre layup and curing; monocoque validated by tensile tests.

CERTIFICATIONS

Design of Industrial Piping Systems 2026
L&T EduTech / Coursera · pipe hydraulics, ASME B31 · verify

Claude Code 101 2026
Claude Academy (Anthropic) · completion badge · verify

AI for Design and Optimization 2026
University of Michigan / Coursera · verify

STAR-CCM+: Fluid Flow & Thermal Simulation 2026
Coursera · verify

Claude Code in Action 2026
Claude Academy (Anthropic) · completion badge · verify

BIM Fundamentals for Engineers 2026
L&T EduTech / Coursera · verify

Computational Fluid Mechanics 2026
Coursera project · airflow around a spoiler · verify

Python for Data Science, AI & Development 2026
IBM / Coursera · verify

SimOps Fundamentals 2025
Simulation process automation